

OUTPUT

```
===== AI Assignment =====
1. Emergency Medicine Delivery (A*)
2. Traffic Signal Optimization
3. Mars Rover Navigation
4. Exam Timetable (CSP)
5. Game AI (Minimax + Alpha-Beta)
6. Exit
Enter Choice: 1
```

```
A* Search
Path: ['A', 'B', 'D', 'G']
Cost: 9
```

```
===== AI Assignment =====
1. Emergency Medicine Delivery (A*)
2. Traffic Signal Optimization
3. Mars Rover Navigation
4. Exam Timetable (CSP)
5. Game AI (Minimax + Alpha-Beta)
6. Exit
Enter Choice: 2
```

```
Hill Climbing
Best Signal Timing: 20
Score: 100
```

```
===== AI Assignment =====
1. Emergency Medicine Delivery (A*)
2. Traffic Signal Optimization
3. Mars Rover Navigation
4. Exam Timetable (CSP)
5. Game AI (Minimax + Alpha-Beta)
6. Exit
Enter Choice: 3
```

```
Mars Rover Navigation
Current Position: (0, 0)
Current Position: (0, 1)
Current Position: (0, 2)
Current Position: (1, 2)
Goal Reached: (2, 2)
```

```
===== AI Assignment =====
1. Emergency Medicine Delivery (A*)
2. Traffic Signal Optimization
3. Mars Rover Navigation
4. Exam Timetable (CSP)
5. Game AI (Minimax + Alpha-Beta)
6. Exit
```

Enter Choice: 4

Exam Timetable

Math -> Day1

Physics -> Day2

Chemistry -> Day3

===== AI Assignment =====

1. Emergency Medicine Delivery (A*)
2. Traffic Signal Optimization
3. Mars Rover Navigation
4. Exam Timetable (CSP)
5. Game AI (Minimax + Alpha-Beta)
6. Exit

Enter Choice: 5

Game AI

Best Value: 5

===== AI Assignment =====

1. Emergency Medicine Delivery (A*)
2. Traffic Signal Optimization
3. Mars Rover Navigation
4. Exam Timetable (CSP)
5. Game AI (Minimax + Alpha-Beta)
6. Exit

Enter Choice: 6

Program Ended

|